

COURSE CONTENTS

Category	Title	Code	Credits-4C			Theory Paper
DID-2	Computer System Organization	CS/IT/EC 401	L	T	P	Max. Marks-100 Min.Marks-35 Duration-3hrs.
			3	1	0	

Unit-I Computer Basics and CPU: Von Newman model, various subsystems, CPU, Memory, I/O, System Bus, CPU and Memory registers, Program Counter, Accumulator, Instruction register, Micro operations, Register Transfer Language, Instruction Fetch, decode and execution, data movement and manipulation, Instruction formats and addressing modes of basic computer. 8085 microprocessor organization

Unit-II Control Unit Organization: Hardwired control unit, Micro and nano programmed control unit, Control Memory, Address Sequencing, Micro Instruction formats, Micro program sequencer, Microprogramming,
Arithmetic and Logic Unit: Arithmetic Processor, Addition, subtraction, multiplication and division, Floating point and decimal arithmetic and arithmetic units, design of arithmetic unit.

Unit-III Input Output Organization: Modes of data transfer – program controlled, interrupt driven and direct memory access, Interrupt structures, I/O Interface, Asynchronous data transfer, I/O processor, 8085 I/O structure, 8085 instruction set and basic programming. Data transfer – Serial / parallel, synchronous/asynchronous, simplex/half duplex and full duplex.

Unit-IV Memory organization: Memory Maps, Memory Hierarchy, Cache Memory - Organization and mappings. Associative memory, Virtual memory, Memory Management Hardware.

Unit-V Multiprocessors: Pipeline and Vector processing, Instruction and arithmetic pipelines, Vector and array processors, Interconnection structure and inter-processor communication.

References:

1. Morris Mano: Computer System Architecture, PHI.
2. Gaonkar: Microprocessor Architecture, Programming, Applications with 8085; Penram Int. Publisher
3. William Stallings: Computer Organization and Architecture, PHI
4. Carl Hamacher: Computer Organization, TMH
5. Tanenbaum: Structured Computer Organization, Pearson Education

COURSE CONTENTS

Category	Title	Code	Credit-4			Theory paper
DID-3	Control System	EC-- 402	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	0	

Unit-I Basic Control System Terminology and Classification of control System, Examples of control System, Transfer Function of Linear Control System, Block Diagram Representation, Signal flow Graph Techniques.

Mathematical Modeling of Electrical Network: AC and DC Servomotors, Error Detector, Stepper Motor, Optical Encoder, Linearization.

Unit-II Sensitivity of Control Systems, Effects of Feedback on gain and time constant, pole location, bandwidth, Sensitivity, Stability, Disturbance signal, Control over System Dynamics by use of Feedback. **Time Response Analysis-** Standard Test Signals, Time Response of 1st Order System, Model of Prototype DC Position Control System, Time Response of Prototype 2nd Order System, Performance Specification of 2nd Order System, Steady-State Errors and Error Constants, Effects of Additions of Poles and Zeros to Open Loop and Closed Loop System, Design Specification of 2nd Order System and Higher-Order System, Performance Indices, Optimal Control System,

Unit-III Time Domain Stability Analysis- Concept of Stability of Linear Systems, Effects of Location of Poles on Stability, Necessary Conditions for Stability, Routh-Hurwitz Stability Criteria, Relative Stability Analysis, Root Locus Concept, Guidelines for Sketching Root-Locus. Frequency Domain Stability Analysis- Performance Specification in Frequency Domain, Co-relation between frequency Domain and Time Domain, Bode Plot, Minimum-Phase and Non-Minimum Phase System, Polar Plots, Inverse Polar Plot, Nyquist Stability Criterion, Assessment of Relative Stability (Phase Margin, Gain Margin and Stability), Constant-M and N Circle, Nichols Chart.

Unit-IV Approaches to System Design, Types of Compensation, Design of Phase-Lag, Phase Lead and Phase Lead-Lag Compensators in Time and Frequency Domain, Proportional, Derivative, Integral and PID Compensation.

Unit-V Concept of State, State Variables and State Model, State Space Representation of Systems, Block Diagram for State Equation, Transfer Function Decomposition, Solution of State Equation, Transfer Matrix, Relationship between State Equation and Transfer Function, Controllability and Observability.

References:

1. B. C Kuo : Automatic Control System, Prentice Hall of India, New Delhi.
2. Nagrath and Gopal : Control System Engineering, New Age International Publishers.
3. Samarjit Ghose : Control Systems Theory and Applications, Pearson Education
4. B. S. Manke : Linear Control System , Khanna Publishers.
5. Ogata : Modern Control Engineering, PHI
6. AK Mandal : An introduction to Control Engineering – Modeling, Analysis and Design, New Age International Publishers

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
Departmental Core DC-3	Digital Electronics	EC-- 403	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	2	

Unit-I Review of Number systems and Binary codes, Binary arithmetic – addition, subtraction, multiplication and division algorithms. **Boolean algebra:** theorems and functions, Simplification of Boolean functions, minimization techniques, Karnaugh's map method, Quine and McCluskey's method, realization of various binary functions using AND ,OR ,NOT,XOR logic gates.

Unit-II Universal gates: NAND, NOR, realization of boolean function using universal gates. Half and full adder, half and full subtractor, Series and parallel adder, BCD adders, look-ahead carry generator. Decoders, Encoders, multiplexers and de-multiplexers. Analysis and design of combination circuits, realization of various Boolean functions using NAND, NOR gates and multiplexers.

Unit-III Multivibrators: Astable, Monostable and bistable multivibrators, 555 timer chip and its application in multivibrators. **Flip-Flops:** R-S, Clocked R-S, T, D, J-K, race around problem, Master-slave J-K., State and Excitation Tables **Shift registers and counters** . synchronous and asynchronous counters, Binary ripple counter, up-down counter, Johnson and ring counter. Analysis and Design of Sequential Circuits.

Unit-IV Semiconductor memories: Organization and construction of RAM, SRAM, DRAM, RAMBUS ROM, PROM, EPROM, EEPROM, PAL and PLAs etc

Unit-V Logic families: RTL, DTL, TTL, ECL, IIL, PMOS, NMOS and CMOS logic etc. Interfacing between TTL and MOS, vice-versa.

References:

1. M. Mano : Digital Logic and Computer Design, Pearson Education
2. W.H. Gothman : Digital Electronics, PHI.
3. Millman and Taub : Pulse, Digital and Switching Waveforms, MGH
4. Salivahanan and Ari Vahagan : Digital Circuits and Design, Vikas Publishing House
5. Leach and Malvino : Digital Principles and Applications, TMH
6. Rajkamal : Digital Systems – Principles and Design, Pearson Education

List of Experiments (Expandable):

All experiments (wherever applicable) should be performed through the following steps.

Step 1: Circuit should be designed/drafted on paper.

Step 2: The designed/drafted circuit should be simulated using simulation Software (TINA-PRO/ PSPICE/ LABVIEW/ CIRCUIT MAKER).

Step 3: The designed/drafted circuit should be tested on the bread board and compare the results with the simulated results.

Step 4: The bread board circuit should be fabricated on PCB by one batch using PCB machine.

1. To test and study of operation of all logic Gates for various IC's.
2. Implementation of AND, OR, NOT, NOR, X-OR and X-NOR Gates by NAND and NOR Universal gates.
3. Binary Addition by Half Adder and Full Adder circuit.
4. Binary Subtraction by Half Subtractor and Full Subtractor circuit.
5. Design a BCD to excess-3 code converter.
6. Verification of the Demorgan's Theorem.
7. Study of RS, JK, T & D flip-flops.
8. Multiplexer/Demultiplexer based boolean function realization.
9. Study and Application of 555 timer (Astable, Monostable, Schmitt trigger, VCO).

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
Departmental Core DC-4	Electronic Circuits	EC--404	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	2	

Unit-I Amplifier Basics, Transistor as an amplifier, load line, Q-point and its selection criteria, designing of fixed bias and self-bias, stability of biasing circuits, calculation of stability factor.

Transistor at low frequency: frequency response, bandwidth, h-parameter analysis of CC, CB and CE configuration, simplified model, gain and impedance calculation of single stage amplifier.

Transistor at high frequency, high frequency model (hybrid- π), Parameters and their definition, Miller capacitance and its effect on voltage gain,

Unit-II Feedback amplifier: positive and negative feedback loop gain, effect of negative feedback on gain stability, distortion, bandwidth, input and output impedance of amplifier, types of feedback (voltage, current, series and shunt) and their analysis.

Oscillators: condition of sustained oscillation, RC phase shift, LC (Hartley and Collpit) Oscillators, Wein Bridge, Negative resistance (Tunnel diode and UJT) oscillators, crystal oscillators.

Unit III Power amplifier, classification, operation, analysis and design of Class A, Class B, Class-AB, Class C, transformer coupled, push pull and complementary symmetry amplifiers, power dissipation in transistors ($P_{d_{max}}$ rating) and efficiency calculations.

Tuned amplifier and its applications, Q factor, selectivity and bandwidth, effect of loading, double tuning (synchronous and stagger)

Unit IV Cascade amplifiers, Calculation of gain, Input and output impedance, Effect of Cascading on bandwidth, Transformer, RC and direct-coupled amplifier and their performance.

Darlington connection, equivalent circuit and Calculation of gain and impedances, Cascade amplifier: advantage, circuit diagram and analysis, feedback pair and applications of BIFET, Bootstrapping technique.

Differential amplifier - configuration, transfer characteristics, DC analysis, h-parameter analysis, differential and common mode gain, CMRR, constant current source and current mirror, level shift.

Unit-V Operational amplifier (IC741), specifications, ideal and practical characteristics, frequency response, unity gain bandwidth, limitations, slew rate and its effect on full power bandwidth, input offset voltage, bias and offset currents, compensation.

Applications of Op-Amp: Inverting and non-inverting amplifier Analog computation, summer (inverting and non-inverting), averager, integrator, differentiator, scalar, sign changer, phase changer, multiplier, buffer, Differential amplifier, instrumentation amplifier, comparator, Schmitt trigger, precision rectifier, log and antilog amplifier, voltage-to-current and current-to-voltage converter.

References:

1. Millman and Halkias : Integrated electronics, TMH

2. Gayakwad ; OPAMP and Linear Integrated Circuits, Pearson Education
3. Boylestad and Nashelsky : Electronic Devices and Circuit Theory, PHI
4. Sendra and Smith : Microelectronics, Oxford Press
5. Graham Bell : Electronic Devices and Circuits , PHI
6. Donald A Neamen : Electronic Circuits Analysis and Design, TMH
7. Salivahanan et al : Electronic Devices and Circuits, TMH

List of Experiments (Expandable):

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Step 1: Circuit should be designed/drafted on paper.

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1. Characteristics of Op-Amp (input offset voltage, slew rate, CMRR, BW, input bias current).
2. Linear application of Op-Amp (voltage follower, inverting and non-inverting amplifier and their frequency response, adder, subtractor, differential amplifier, integrator and differential frequency response)
3. To design and construct a shunt and series regulator and find line and load regulation.
4. Design and performance evaluation of transistor amplifiers in CE, CB and CC configuration
5. Design and performance evaluation of FET amplifiers.

COURSE CONTENTS

Category	Title	Code	Credit-6			Theory paper
Departmental Core DC-5	Analog Communication	EC-405	L	T	P	Max. Marks-100 Min. Marks: 35 Duration: 3 hrs.
			3	1	2	

Unit-I Different types of Signals (Continuous, Discrete, Periodic), Time Domain and Frequency Domain Representation, Introduction to basic Transform Techniques applicable to these Signals. **Spectral Analysis:** Fourier Technique, Fourier Transform and their Properties, Transform of Gate Signal, Impulse Function and Unit Step Function, Fourier Transform Technique for Periodic Signal, Transform of Train of Pulses and Impulses, Sine and Cosine wave.

Signal Energy and Power, Spectral Density of various types of signals, Spectra (Parseval's Theorem), Density Spectra of Periodic Gate and Impulse train.

Linear Time Invariant (LTI) Systems, Impulse Response, Convolution, Convolution with Impulse Function, Casual and Non Casual System, Distortion less System, Impulse Response of Distortion less System, Ideal Filter and Practical Filter.

Unit-II Modulation Techniques: Need and types of modulation techniques, Amplitude Modulation, Frequency Spectrum, Power Distribution, Modulation by Complex Signal, Low Level and High Level AM Modulators, Linear Integrated Circuit AM Modulators, Suppressed Carrier Generation (Balance/Chopper and Square Law Modulation), SSB Generator (Phase and Frequency Discrimination Method), VSB Transmission and Application. Detection of AM signals: Envelope Detector Circuit, RC Time Constant, Synchronous Detection Technique, Error in Synchronous Detection, SSB signal detection, PLL and its use in demodulation.

Unit-III Angle Modulation: Frequency and Phase Modulation Frequency spectrum, bandwidth requirement, Frequency and Phase Deviation, Modulation Index, NBFM and WBFM, Multiple frequencies FM. FM Modulators: Direct (Parameter Variation Method) and Indirect (Armstrong) Method of frequency modulation. FM Detector: Slope Detector, Foster Seely Discriminator, Ratio Detector and PLL detectors.

Unit-IV Radio Transmitters: AM transmitter, block diagram and working of Low Level and High Level Transmitters, Trapezoidal Pattern and Carrier Shift, SSB Transmitters, FM transmitters - Frequency Multiplication Applied to FM Signals, FM transmitters.

Radio Receivers: Block Diagram of Radio Receiver, Receiver Characteristics (Selectivity, Fidelity and Sensitivity), AM Receiver, RF Receiver, Super-heterodyne Receiver, RF Amplifier, Frequency Mixer, AVC and AFC, Image Signal, Intermediate Frequency Selection, Diversity Reception, FM Receiver.

Unit-V Noise : Sources and types of noise and their power density, White Noise, Noise from Single and Multiple noise source for Linear Systems, Super Position of Power Spectrum, Equivalent Noise Bandwidth, Noise Figure, and Equivalent Noise Temperature, their Relationship, Calculation of Noise Figure and Noise Temperature for Cascade Systems, Noise Performance of Communication System, Band Pass Noise Representation in Terms of Low Pass, In-phase and Quadrature Phase Component and their Power Spectral Density, Figure of Merit, Calculation for AM, AM-SC and SSB System, Noise in Angle Modulated System, Figure of Merit for FM, Noise Density of Output of FM

Detector, Pre-Emphasis and De-Emphasis, Phasor Representation of Noise, Capture Effect, Comparison of Noise Performance of AM and FM.

References:

1. B.P. Lathi : Modern Analog and Digital Communication System, Wiley Eastern limited
2. Taub and Schilling : Principles of communication Systems, TMH
3. Singh and Sapre : Communication Systems, TMH
4. S Haykin : Communication Systems, John Wiley and Sons Inc
5. S Ghose: Signals and Systems, Pearson Education.
6. A Bruce Carlson : Communication System, TMH
7. Steven : Communication Systems – Analysis and Design, Pearson Education

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1. Analysis of AM Modulation and Demodulation Techniques (Transmitter and Receiver), Calculation of Parameters
2. Analysis of FM Modulation and Demodulation (Transmitter and Receiver) and Calculation of Parameters
3. To Construct and Verify Pre-emphasis and De-emphasis and Plot the Waveforms.
4. Study of Super-heterodyne Receiver and Characteristics of Radio Receiver.
5. To Construct Frequency Multiplier Circuit and to Observe the Waveform
6. Study of AVC and AFC.
7. Study of PLL chip (566) and its use in various systems

COURSE CONTENTS

Category	Title	Code	Credit-4			Practical
IT-3	Software Lab I	EC- 406	L	T	P	Max. Marks-50 Min. Marks: 25 Duration: 3 hrs.
			-	-	4	

SECTION - A MatLab / SciLab / Similar software (Mathematical/Scientific Laboratory Software) (SciLab, a freeware can be downloaded from www.scilab.org)

Introduction to Scilab / MatLab, Study of Scilab / MatLab programming environment, Modeling, Design and development of Programs.

Programs Related to Analog Communication- (Example-Plots of Different Signals and their Fourier Transforms, Computation of Linear and Cyclic Convolution between Two Signals, Simulation of Different Types of Amplitude transmitters and Receivers, AM Transmitter and Receiver, FM Transmitter and Receiver, Simulation of a Communication System (Generation, addition of noise and Detection).

Programs Related to Control System- Open-Loop and Closed Loop Control System Response using Scilab/ MatLab, Determining Transient Response, Specification of Second Order System, Effect of PID controller on Control System, Bode Plot, Nyquist Plot and Root Locus Plot.

SECTION-B CIRCUIT SIMULATION/ PCB DESIGNING SOFTWARES

Study of Circuit Simulation Software (any one - TINA-PRO/ PSPICE/ LABVIEW/ CIRCUIT MAKER). PCB Layout Software (any one - PROTEL/ ORCADE/ ALTERA).

Design and Simulation of basic Electronic Circuits (Example Rectifiers, Amplifiers, Oscillators, Digital Circuits, Transient and steady state analysis of RC/RL/RLC circuits etc)

References:

1. Vinu V. Das: Programming in Scilab, New Age Publisher.
2. Chapman Stephen J.: MATLAB Programming for Engineers, 3rd Edition, Thomson Cengage
3. Rudra Pratap: Getting Started with MATLAB 7, Oxford University Press (Indian Edition).
4. Proakis John G.: Contemporary Communication System Using MATLAB; Thomson Vikas Pub.
5. B.S. Manke: Linear Control Systems - with MATLAB Application, Khanna Publishers.
6. Simulation/Designing Software Manuals.
7. S. Hassan Saeed : Automatic Control Systems (with MATLAB Programming) S. K. Kataria and Sons New Delhi.

List of Experiments:

Programs to be performed based on the topics contained in the syllabus.